VHDL – Lab 7

The master lab

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Revision history

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2 General

2.1 Background

All previous lab assignments have been created to, step by step, introduce new challenges in VHDL design. In this assignment you will need to use all your skills and learn how a real project can be developed.

2.2 Intended outcome

In this lab the goal is to understand how a larger VHDL design can be specified and developed. The important concept of a well written testbench is again demonstrated.

2.3 Checkpoints

In order to Pass the lab assignment the student shall be able to show that the following tasks have been fulfilled.

- Properly written VHDL implementation. Use proper indentation, spaces are cheap! Add comments when you feel it's suitable.
- The design shall be implemented according to the Design specification.
- The design shall pass all testbench tests.
- The design shall be properly pin-planned and pass timing analysis.
- The worst case timing path for setup-requirement shall be checked and identified in code.
- The design shall pass all synthesis and Place & Route steps without any critical or strange warnings that can be solved easily.
- Design shall also be tested in hardware preferably with control from both key push buttons and Serial connection.

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3 Lab instructions

3.1 General

The mission is to implement a PWM function according to a design specification. The design specification specifies how the design shall work and how the design shall be tested using a testbench.

You will be able to download a few design files from the course homepage, such as the Serial UART component, the PLL component, the reset control component, and help for handling a generic controlled instantiation in the top level.

The testbench that is designed to test the design according to its specification, and a constraints file that shall be used to constrain the clock frequency for timing analysis are also available in the course homepage.

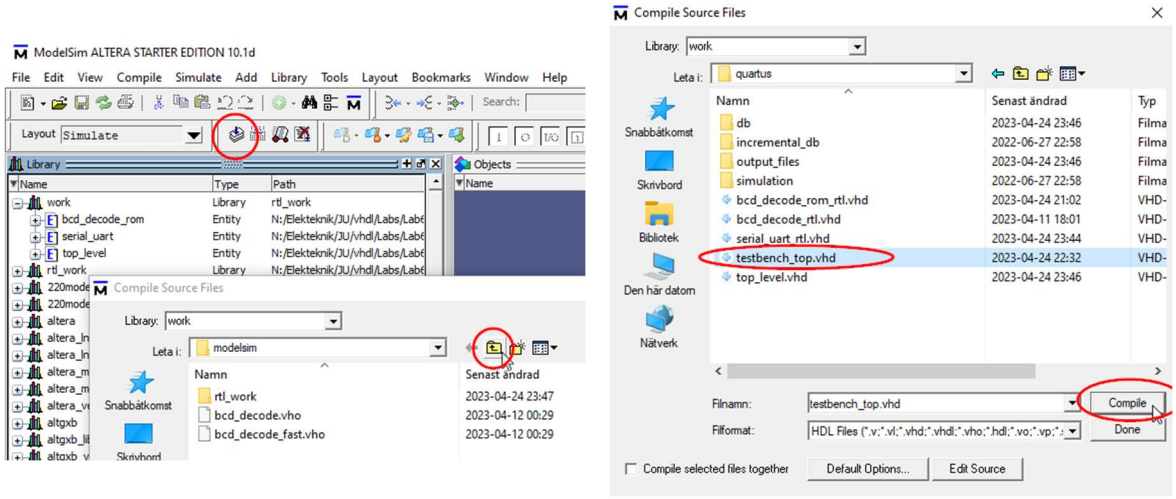
The idea is that when you have designed your design in a way that it completes the testbench it should also work in HW after a correct pin-planning is performed.

Since this assignment is more complex than other labs the suggestion is to team up in groups of **maximum 2 persons** and divide a few components between each other. The specification should be clear enough to understand what each sub component shall do. Ensure you read the design specification and understand the purpose of each sub-component in the design and also ensure that you understand the interface between the sub-components before starting.

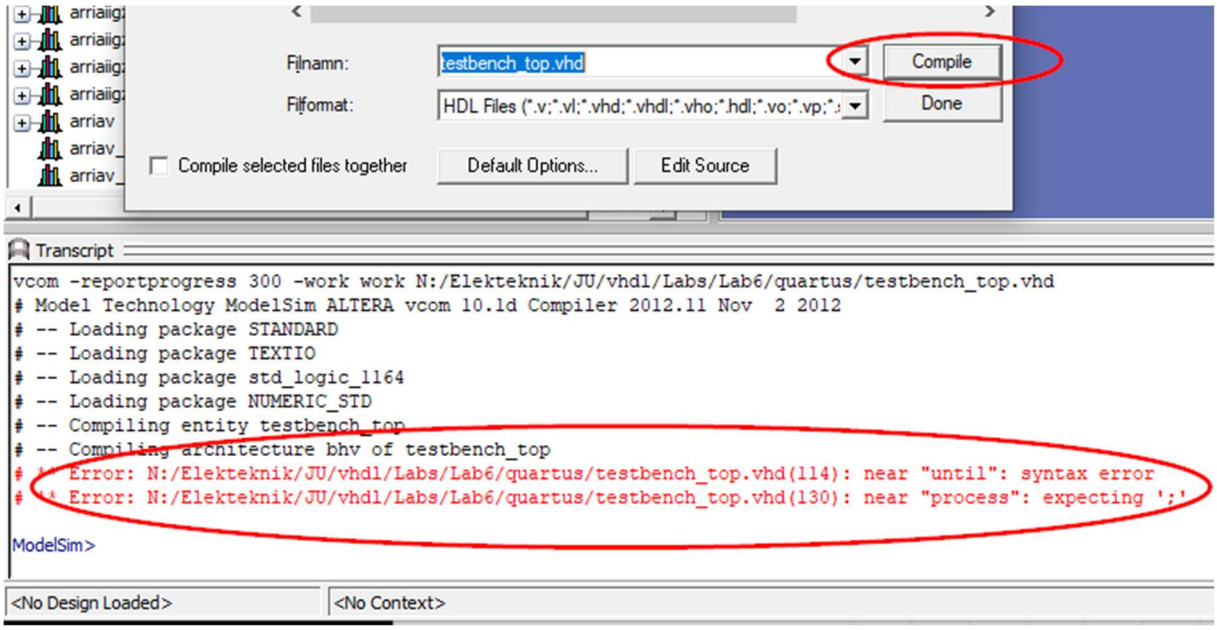
3.2 Running a testbench in Modelsim

To run a testbench in Modelsim use the compile button and compile the testbench.

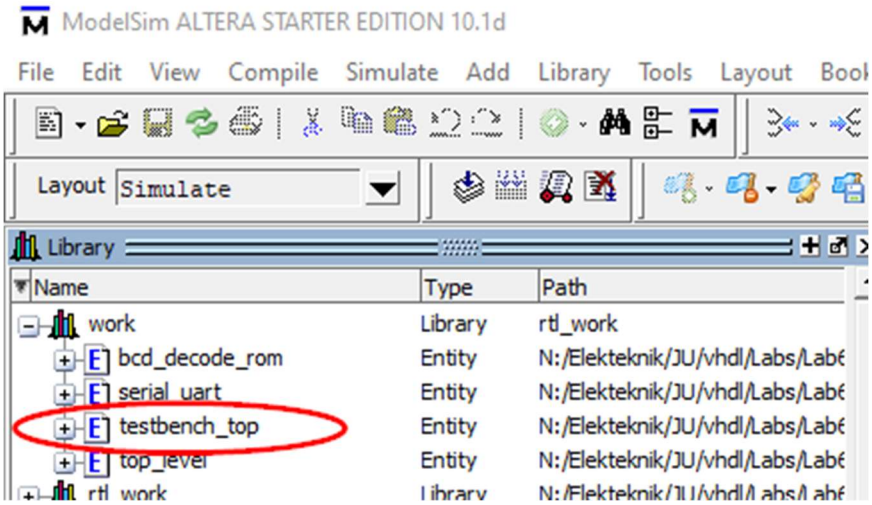
The testbench file is hopefully stored in the project folder so you will need to browse to it from the modelsim working directory, by clicking “up a level” in the folder structure.



Check Modelsim Transcript window to ensure that no compile errors is found.

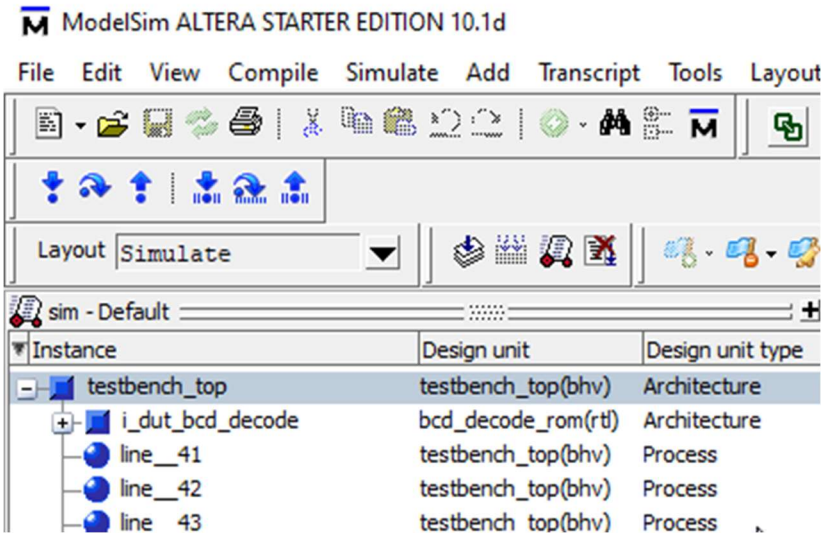


Once the testbench is compiled successfully you should be able to find it in the library “work”.



Double click the testbench top entity to start simulation.

Simulation should start and the DUT should be shown as an instance.



Drag the signals of interest (preferably all of them), as you're used to into the wave window, group them to not mix them up.

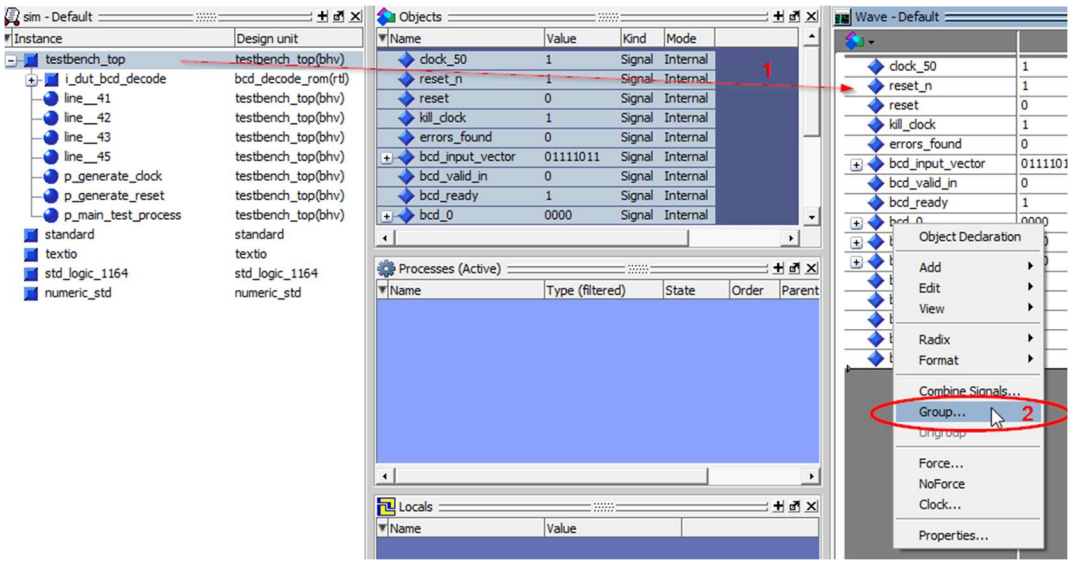
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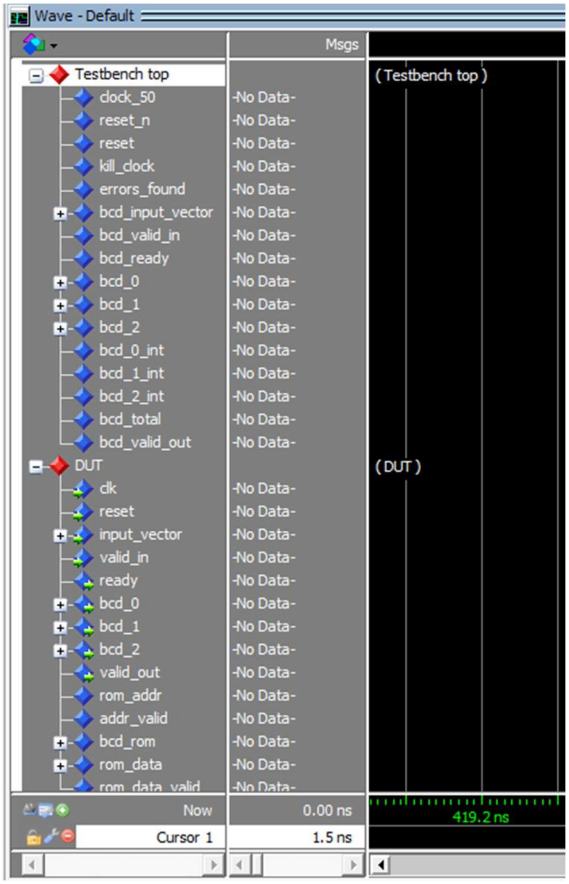
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You group signals by right clicking marked signals in the wave window and selecting “Group…”





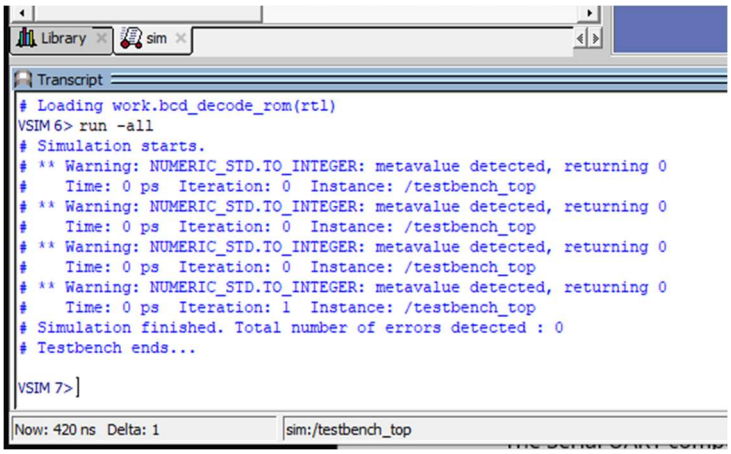
Name the group something smart, e.g. instance name, and repeat for all instances.

Run the full simulation with the command “run -all”

This command will run the testbench and make modelsim stop simulation when all clocks and sequential processes are stopped (see kill_clock signal in testbench file).

You will probably get a few warnings early in simulation, **this is normal and can safely be ignored.**

The transcript window should hopefully look something similar to the screenshot below:



Wave window should be updated and look similar to below:

